SA-Sur : Note to Frequency

User manual

Applies to SA-Sur 1.2.0 and later. Issue 2, 20 October 2026.

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1. About SA-Sur

SA-Sur is a single-window app that shows the frequency of a note and plays it as a sine tone.

It covers every note of a twelve-tone equal temperament scale, from C0 to B8.

SA-Sur is free of charge. **Hafod Mastering Limited** and **Sudeep Audio.com LLP** publish it together, for macOS and Windows.

About the name: *SA* is the first note of the Indian scale, and also the abbreviation of Sudeep Audio. *Sur* is note, or pitch.

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2. Install SA-Sur

macOS - the download version

Requirements: macOS 13.0 or later, Apple Silicon or Intel.

1. Download `SA-Sur-1.2.0-MacOS.zip` from the Releases page.
2. Double-click the zip file. macOS unpacks the folder.
3. Move `SA-Sur.app` to the Applications folder.
4. Open SA-Sur.

The app is signed and notarized by Apple. macOS shows the usual confirmation for a downloaded app, and no other message.

macOS - the App Store version

1. Open the App Store.
2. Search for `SA-Sur`.
3. Click *Get*, then click *Install*.

The App Store copy is the same app and the same version, sandboxed for the store. The App Store updates it for you.

Windows - the download version

Requirements: Windows 10 version 1809 or later, 64-bit.

1. Download `SA-Sur-1.2.0-windows.exe` from the Releases page.
2. Run the file.

There is no installer, and no .NET runtime to add.

The release download is unsigned. Windows therefore shows "*Windows protected your PC*" and the publisher **Unknown publisher** on the first run. Do this:

1. Click *More info*.
2. Click *Run anyway*.

Windows - the Microsoft Store version

1. Open the Microsoft Store.
2. Search for `SA-Sur`.
3. Click *Get*, then click *Install*.

The Store copy is the same app and the same version. It is signed by Microsoft, so it shows no unknown-publisher warning, and the Store updates it for you.

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3. The window

All the controls are in one window. The window keeps its size.

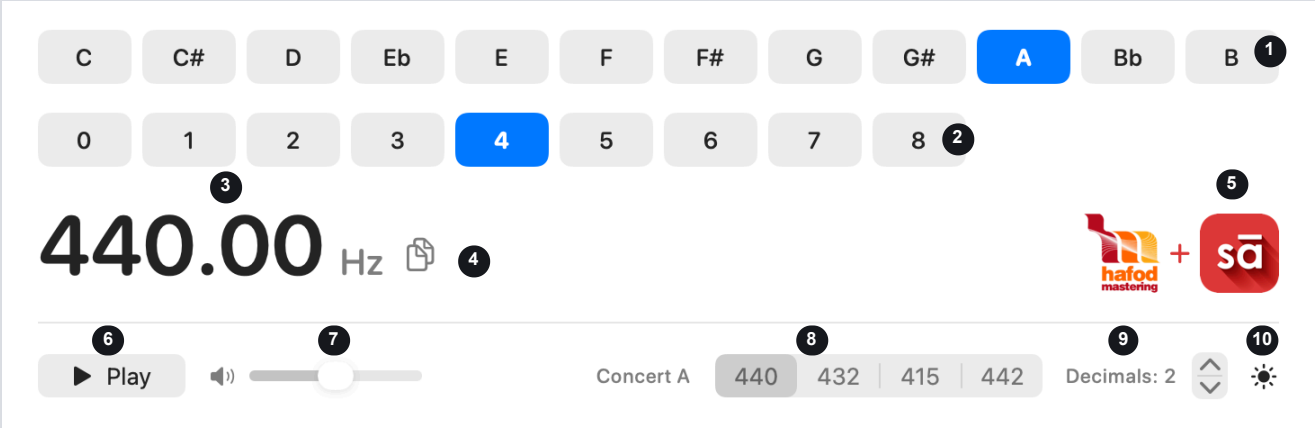


Figure 1 - the parts of the SA-Sur window.

#	Item	Where	What it does
1	Note buttons	top row	Select the note: `C C# D Eb E F F# G G# A Bb B`
2	Octave buttons	second row	Select the octave: `0` to `8`
3	Readout	below the octave row, large digits	Shows the frequency in hertz
4	Copy icon	right of the readout	Copies the frequency
5	Company marks	right of the octave row	Open the Hafod Mastering website or the Sudeep Audio website
6	Play / Stop	bottom left	Starts and stops the tone
7	Level slider	next to the Play button	Sets the output level
8	Concert A	bottom right, first group	Sets the concert-A reference: `440`, `432`, `415`, `442`
9	Decimals	bottom right, after Concert A	Sets the number of decimal places: 0 to 6

#	Item	Where	What it does
10	Appearance	bottom right, last icon	Selects the System, Light or Dark appearance

SA-Sur opens in the state A4, octave 4, 440 Hz and 2 decimal places. The readout is therefore 440.00 Hz.

The macOS app also has a **Help** menu in the menu bar. The Windows app has no menu bar, so it carries the same items behind a small Help icon at the right edge of the bottom row. Section 4 describes both.

SA-Sur quits when you close the window.

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4. Procedures

Select a note

1. Find the note in the top row.
2. Click the note.

SA-Sur shows the frequency of the note. If a tone is playing, SA-Sur retunes the tone at once.

Select an octave

1. Find the octave in the second row.
2. Click the octave.

SA-Sur shows the new frequency. If a tone is playing, SA-Sur retunes the tone at once.

Play and stop the tone

1. Use the Play button, or press the **Space bar**.
2. Use the same control again to stop the tone.

The button shows `Stop` while the tone plays.

The Space bar works wherever the focus is in the window, so it still plays after you click a note.

Set the level

- Move the slider next to the Play button.

The slider opens at 50%. The output level is limited to 11% of full scale, so a full slider is not loud. Section 6 gives the details.

Set the concert-A reference

1. Click `440`, `432`, `415` or `442` in the Concert A group.
2. Look at the readout.

SA-Sur changes the readout and the tone. If a tone is playing, SA-Sur retunes the tone at once. `440` Hz is the usual reference. `432`, `415` and `442` Hz are other references in common use.

Set the number of decimal places

1. Click the up arrow or the down arrow of the Decimals stepper.

2. Look at the readout.

The range is 0 to 6 decimal places. At 6 decimal places, A4 shows 440.000000.

Copy the frequency

1. Click the copy icon.
2. Paste the value into another application.

SA-Sur puts the number on the clipboard, without the unit Hz. The icon shows a check mark for about one second.

Change the appearance

1. Click the appearance icon at the bottom right.
2. Click it again to see the next appearance.

The icon goes through System, Light and Dark. SA-Sur keeps your choice for the next time that you open the app. On Windows the title bar follows the choice as well.

Open the manual, the licences or the version

1. Click the Help icon at the bottom right, or press **F1**.
2. Choose *User Manual*, *Licences and notices*, or *About SA-Sur*.

On macOS the same items are in the **Help** menu. The manual opens as a PDF. The licences window shows the SA-Sur licence, the Apache-2.0 text, the notices for the third-party components, and the font licence.

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5. Frequency values

SA-Sur calculates each frequency from the concert-A reference. The frequency doubles at each octave, and there are twelve equal steps between two octaves.

Note	Frequency
C0	16.35 Hz
A3	220.00 Hz
A4	440.00 Hz
A5	880.00 Hz
B8	7902.13 Hz

The lowest note is C0 at 16.35 Hz. The highest note is B8 at 7902.13 Hz.

Rounding, and printed charts that disagree

Some printed charts give F#2 as 93 Hz and G0 as 25 Hz. The correct values are 92.4986 Hz and 24.4997 Hz. To two decimal places these values are 92.50 Hz and 24.50 Hz, and to whole numbers they are 92 Hz and 24 Hz.

A chart that rounds an intermediate value first shows a value one digit high. SA-Sur calculates the frequency directly from the reference, so its readout is correct where such a chart is not.

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6. The tone

SA-Sur plays a sine tone. A sine tone has one frequency and no harmonics.

The level goes up and down over 12 ms. Therefore, you hear no clicks when the tone starts, when the tone stops, or when the pitch changes.

The level follows the slider. The maximum level is 11% of full scale.

> **WARNING:** Do not use a high level with headphones. A sine tone at a high frequency is more tiring to the ear than music at the same level. Set the level low first, then increase it if necessary.

SA-Sur sends the tone to the default output device of the computer. To use a different device, change the output device in the sound settings of the computer:

- **macOS:** System Settings > Sound.
- **Windows:** Settings > System > Sound.

SA-Sur follows the new default device and continues to play. If the device disappears while the tone is playing, press Play again when another device is available.

SA-Sur keeps the audio engine ready after the first time that you play the tone. The next start is therefore immediate. SA-Sur releases the audio device when you close the window.

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7. If you have a problem

There is no sound

1. Increase the level with the slider.
2. Make sure that the computer uses the correct output device, and that its volume is not zero.
3. Press the Space bar to stop the tone. Press it again to start the tone.

SA-Sur says that it cannot start the audio

On macOS, SA-Sur shows a message when the audio engine refuses to start: for example when there is no output device at all. Connect an output device, or choose one in System Settings > Sound, then press Play again.

On Windows, the reason appears in red beside the readout, for example *No audio output device*.

macOS will not open a build that you made yourself

A build from the source code is not notarized. macOS therefore stops the first start. Do this:

1. Try to open SA-Sur.
2. Dismiss the message.
3. Open System Settings.
4. Go to Privacy & Security.

5. Click *Open Anyway* for SA-Sur, then confirm.

For advanced users: in Terminal, run this command.

```
xattr -dr com.apple.quarantine /Applications/SA-Sur.app
```

The command removes the quarantine flag from the app. No administrator password is necessary.

Windows warns about an unknown publisher

The release download is unsigned. Section 2 gives the two steps. The Microsoft Store version is signed by Microsoft and shows no such warning.

The App Store version will not open

Reinstall the app from the App Store. This corrects most install problems.

There is a security problem

Read `SECURITY.md` in the repository.

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8. Privacy

SA-Sur makes no network connections. It collects no data and it sends no data. There is no analytics, no telemetry, no advertising and no tracking SDK. The macOS build uses only Apple frameworks; the Windows build uses open-source UI and audio libraries (Avalonia, NAudio) that make no network connections.

SA-Sur stores one setting on the computer: the appearance choice (System, Light or Dark).

The window shows the marks of the two companies. A click on a mark asks your browser to open that company's website. That visit is a usual visit to a website, and the privacy terms of that website apply.

The full policy is in `docs/PRIVACY.md` in the repository.

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9. Licence and marks

The **source code** of SA-Sur is released under the **Apache License 2.0**. You can use it, change it and redistribute it, also inside a product whose source is closed. Keep the licence notices, and state the significant changes that you make.

A copy bought from the App Store or the Microsoft Store is licensed under the **SA-Sur End User Licence Agreement**. That agreement grants you the store copy, and it also carries the Apache-2.0 grant for the source: your rights in the source are never narrower than Apache-2.0, whichever copy you installed.

The Windows build includes third-party components under the MIT licence and the Nunito typeface under the SIL Open Font License. Their notices are in `THIRD-PARTY-NOTICES.md` and in the *Licences* item of the Help menu.

There is no paid edition and no support contract.

The names, the logos and the marks of Hafod Mastering and Sudeep Audio, and the **SA-Sur** name and app artwork, are not part of the code licence (Apache-2.0, section 6). Forks are welcome. Replace those marks, or remove them, from a build that you publish.

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10. Help and links

- Source code and documentation: https://github.com/hafodmastering/sa-sur
- End user licence: https://github.com/hafodmastering/sa-sur/blob/main/EULA.md
- Privacy policy: https://github.com/hafodmastering/sa-sur/blob/main/docs/PRIVACY.md
- Hafod Mastering: https://hafodmastering.co.uk
- Sudeep Audio: https://sudeepaudio.com

Support

Neither company supplies product support for SA-Sur. Neither company reviews changes or collects feature requests. The app is published free of charge, as it is.

The source code is open and short. Give it to an AI assistant that you already use, and ask for the change that you want. That is the fastest route to the app that you need.

Appendix A - note frequencies at A4 = 440 Hz

Values are given to four decimal places. For another concert-A reference, multiply the value by the reference divided by 440. Example: at 432 Hz, A4 is 432.0000 Hz.

Note	Hz	Note	Hz	Note	Hz
C0	16.3516	C3	130.8128	C6	1046.5023
C#0	17.3239	C#3	138.5913	C#6	1108.7305
D0	18.3540	D3	146.8324	D6	1174.6591
Eb0	19.4454	Eb3	155.5635	Eb6	1244.5079
E0	20.6017	E3	164.8138	E6	1318.5102
F0	21.8268	F3	174.6141	F6	1396.9129
F#0	23.1247	F#3	184.9972	F#6	1479.9777
G0	24.4997	G3	195.9977	G6	1567.9817
G#0	25.9565	G#3	207.6523	G#6	1661.2188
A0	27.5000	A3	220.0000	A6	1760.0000
Bb0	29.1352	Bb3	233.0819	Bb6	1864.6550
B0	30.8677	B3	246.9417	B6	1975.5332
C1	32.7032	C4	261.6256	C7	2093.0045
C#1	34.6478	C#4	277.1826	C#7	2217.4610
D1	36.7081	D4	293.6648	D7	2349.3181
Eb1	38.8909	Eb4	311.1270	Eb7	2489.0159
E1	41.2034	E4	329.6276	E7	2637.0205
F1	43.6535	F4	349.2282	F7	2793.8259
F#1	46.2493	F#4	369.9944	F#7	2959.9554
G1	48.9994	G4	391.9954	G7	3135.9635
G#1	51.9131	G#4	415.3047	G#7	3322.4376
A1	55.0000	A4	440.0000	A7	3520.0000
Bb1	58.2705	Bb4	466.1638	Bb7	3729.3101
B1	61.7354	B4	493.8833	B7	3951.0664
C2	65.4064	C5	523.2511	C8	4186.0090
C#2	69.2957	C#5	554.3653	C#8	4434.9221
D2	73.4162	D5	587.3295	D8	4698.6363
Eb2	77.7817	Eb5	622.2540	Eb8	4978.0317
E2	82.4069	E5	659.2551	E8	5274.0409
F2	87.3071	F5	698.4565	F8	5587.6517
F#2	92.4986	F#5	739.9888	F#8	5919.9108
G2	97.9989	G5	783.9909	G8	6271.9270
G#2	103.8262	G#5	830.6094	G#8	6644.8752
A2	110.0000	A5	880.0000	A8	7040.0000
Bb2	116.5409	Bb5	932.3275	Bb8	7458.6202
B2	123.4708	B5	987.7666	B8	7902.1328